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The Reinforcement Learning Contour

From Document Production to Governed Epistemic Transformation in AI-Assisted Scientific Inquiry

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THE PROBLEM

Cheap text is not cumulative knowledge

Generative AI has sharply reduced the technical cost of producing scientific text. But higher production capacity does not, by itself, entail cumulative knowledge, conceptual novelty, or epistemic progress.



Textual novelty without epistemic novelty A new title or vocabulary is not a new claim.



Closed, self-confirming corpus A corpus grows coherent yet decreasingly corrigible.



Unverified external expansion Bias enters as unchecked novelty; citations fabricated.



Document treated as an endpoint The artifact ends the episode instead of modifying it.

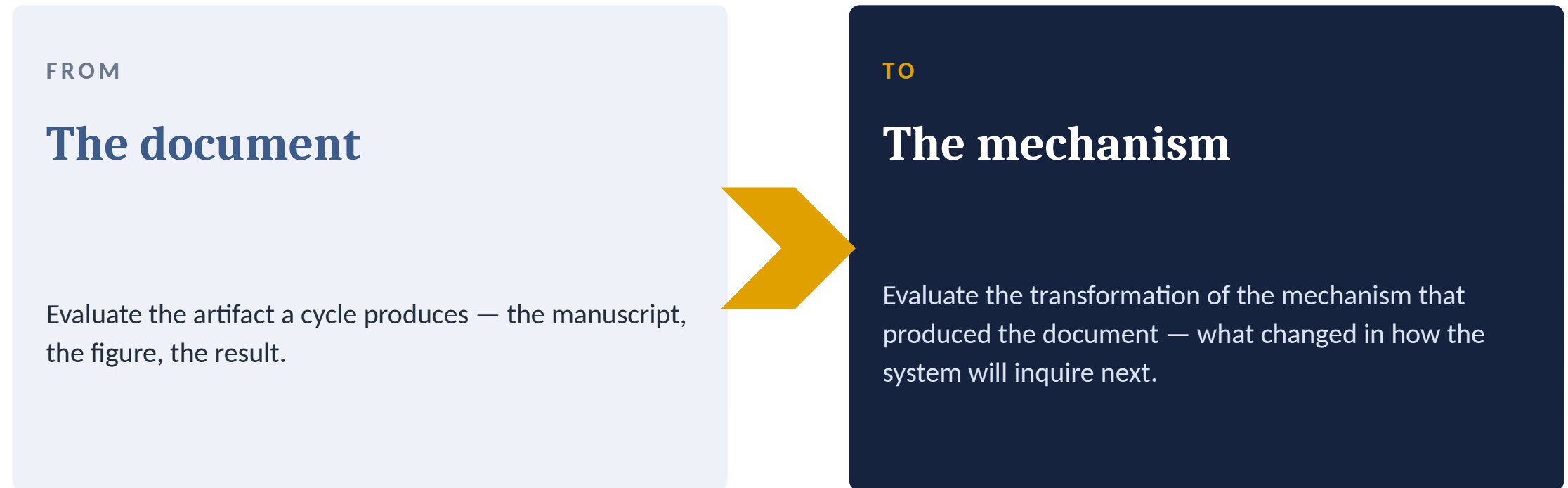
PRODUCTION



PROGRESS

The right question is not how to automate documents, but how each research episode beneficially modifies the mechanism of inquiry.

A new unit of scientific analysis



A research cycle can be epistemically successful even when it produces no publishable document — if it improves the mechanism from which future inquiry emerges.

Epistemic contour vs. the RLC

Epistemic contour C_t

The time-indexed state of a bounded knowledge-action configuration — what the system knows and how it perceives, questions, and evaluates at moment t .

$$C_t = \langle K, O, G, R, P, E, M, H \rangle$$

knowledge · ontology · graph · retrieval · policy · criteria · memory · governance

Reinforcement Learning Contour

The governed recursive architecture that links contour states through evidence, action, evaluation, approval, and reintegration. “Reinforcement” lives in the mechanism of transition — not in the static state.

$C_t \xrightarrow{[X_t]} C_{t+1}$ a research episode X_t yields not just a document, but a governed transition between contours.

Not a reinforcement-learning algorithm

The name is retained for the recursive principle — evaluated outcomes modify the conditions and policies governing future action. The relationship to technical RL is analogical, stated precisely in the paper.

- ✗ **No Markov decision process** The state space is open and partly unformalized.
- ✗ **No scalar reward** Evaluative criteria are multidimensional and contested.
- ✗ **No value function / policy optimization** Updates include human judgment, not return maximization.
- ✗ **RL concepts apply as warnings** Reward hacking is a live hazard the design guards against.

The Law of Experiential Value

The greater the beneficial, traceable differentiation between a preceding epistemic contour and the contours governing future perception, decision, and action — the greater the value of the research cycle.

$$V(X_t) \propto D_+(C_t, C_{t+1})$$

Beneficial, not merely different. Any episode alters a system; alteration may produce confusion, false confidence, or degraded calibration. D_+ is a relational judgment supported by evidence and governance records — never a scalar reward to be optimized.

This reframes the unit of analysis from the document to the transformation of the mechanism that produced it.

Weak learning vs. strong learning

Weak learning

$$A_t \rightarrow A_{t+1}$$

Changing an answer. The system produces a different output, but the function that produces answers is unchanged.

Strong learning

$$f_t(\cdot) \rightarrow f_{t+1}(\cdot)$$

Changing the function itself — a restatement of double-loop learning. An auditable claim of strong learning requires snapshots, differences recorded with provenance, and acceptance separated from generation.

Standing on four traditions

1

Popper

Criticism & exposure to refutation — operationalized as an adversarial stage, not a score.

2

Lakatos

Progressive vs. degenerating research programmes — made contemporaneous and inspectable.

3

Kuhn · Laudan · Longino

Progress is multidimensional; objectivity is a property of critical interaction among situated inquirers.

4

Argyris & Schön

Single-loop vs. double-loop learning — instrumented with snapshots, provenance, and authority.

Governance is who may commit

Prior Research Corpus	may propose
Locally Governed Knowledge Store	may propose
Temporal Provenance & Relation Layer	may propose
Source-Grounded Synthesis	may propose
Adversarial External Discovery	may propose
Research Orchestration & Drafting	may propose
Human Epistemic Governance	SOLE ACCEPTANCE

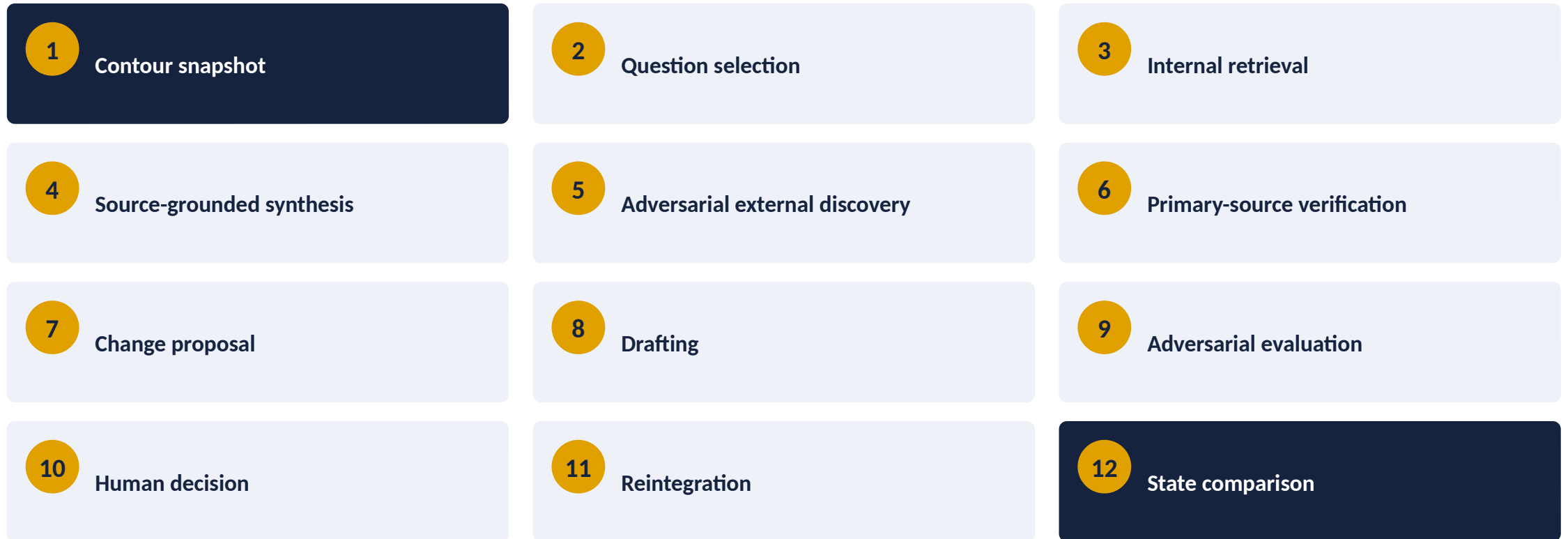
Generation Authority



Acceptance Authority

Roles 1–6 may propose changes to the contour; only the human may commit them. That separation is structural, not procedural.

Twelve steps, two load-bearing



Steps 1 and 12 — snapshot and state comparison — are the load-bearing pair. A cycle run without both is ordinary AI-assisted writing.

Documented, not scored

Claim change

none / refined / rejected / introduced

Evidence change

lower / unchanged / higher / contested

Contradiction

none / unresolved / partial / resolved

Ontology

none / local / structural

Uncertainty

increased / unchanged / reduced*

Inquiry policy

no / possible / demonstrable

Provenance

absent / partial / complete

External review

no / internal only / external

No composite score. Each dimension is documented separately — a categorical record, never a single optimized index. Once a measure becomes a target, novelty and contradiction handling become reward-hacking surfaces.

What can go wrong — and the guardrail

Automated paper inflation	➤	Count governed cycles (incl. justified non-publication), not papers.
Conceptual echo chamber	➤	Mandate adversarial retrieval — ≥ 1 serious competing explanation per cycle.
Citation laundering	➤	Verify the original source; enforce claim-to-source provenance & hierarchy.
Reward hacking	➤	Use no composite score; balance novelty with evidence and falsifiability.
Reflexive self-validation	➤	Separate generation from evaluation; preregistered rubrics, blinded, external audit.

The reflexive validation problem

The RLC contains a structural circularity: the same human-machine environment may generate an epistemic change, define the categories through which it is represented, detect it, and judge whether it is beneficial.

- Separate roles: generation $\neq$ evaluation $\neq$ source verification
- Compare central claims blindly against earlier work
- Keep rejected changes visible; audit ontology & provenance
- Discount any favourable evaluation without independent, blinded assessment — including the author's own

Managed, not solved. Human veto does not eliminate the problem — the human may also be the architect, corpus author, and beneficiary.

Seven testable hypotheses

- H1** Greater claim-level novelty than non-recursive LLM drafting (blinded).
- H2** Fewer unsupported claims and source-to-claim mismatches.
- H3** Temporal provenance improves detection of superseded/contradicted claims.
- H4** Mandatory change records reduce repetition across manuscripts.
- H5** Better confidence calibration than fully automated workflows, at a throughput cost.
- H6** Cycle value correlates with future-question quality, not length or speed.
- H7** Contradictory evidence yields more benefit under governance — more harm without it.

PILOT

30–90 cycles

Three conditions:

- manual research
- AI drafting, no recursion
- full RLC architecture

Blinded expert assessment;
preregistered rubrics; ≥ 2
independent raters.

One local instantiation — and its caveats

A single local implementation is offered as an existence proof, not validation — products are replaceable so long as the authority allocation holds:

Knowledge store

Obsidian

Provenance

Graphiti

Synthesis

NotebookLM

Adversarial discovery

Perplexity

Orchestration

Codex

Final authority

Human researcher

Explicitly an untested conceptual proposal

The pilot has not been run; beneficial differentiation is contestable and often visible only over time; the evaluator is not fully independent of the generator; a one-researcher corpus can preserve that researcher's biases; and daily iteration could itself privilege rapid production over slow validation — the very imbalance the framework opposes.

TAKEAWAY

Restraint is not the absence of productivity.

It may be evidence that the system has learned to distinguish document production from scientific contribution.

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emanuelshirbint.github.io/research/works/reinforcement-learning-contour/